NOVACAP FINANCIAL TECHNOLOGIES

Principal Strategist — 90-Day FinTech Arbitrage Academy

Principal Strategist — Training Program

Master Algorithmic Arbitrage Trading with Python

> *A structured 90-day journey from data ingestion to SOCPA-compliant trading systems. No external dependencies. No shortcuts. Real standards.*

How This Program Works

The Principal Strategist simulator generates **90 daily challenges** across three phases. Each day follows the same cycle:

```
START → Briefing → Code → Run → EOD → Score ≥ 100? → ADVANCE
                ↑                                      ↓
           Fix violations ←── Score < 100 ←────────────┘
```

The rule is simple: score 100/100 on every day's evaluation, or you do not advance. There is no partial credit. There is no "good enough."

Prerequisites

Before you begin, you should be comfortable with:

SKILL	LEVEL REQUIRED	HOW IT'S TESTED
Python syntax	Write functions, loops, conditionals	Day 1 boilerplate
Type annotations	PEP 484 — `def f(x: int) -> str:`	AST auditor checks

SKILL	LEVEL REQUIRED	HOW IT'S TESTED
Exception handling	`try/except/finally` blocks	Critical AST violation if missing
Basic SQL	`SELECT`, `INSERT`, `CREATE TABLE`	Ledger Engine tasks
HTTP concepts	GET/POST, JSON, status codes	Mock Exchange interaction
Command line	Running Python scripts, navigating directories	CLI interaction

If you cannot do all of the above, start with Days 1–3 in "learning mode" (score is informational, not blocking) until you're comfortable.

The Three Phases

Phase 1: Data Ingestion & Foundational Risk Math (Days 1–30)

Theme: You are a data pipeline. Fetch, calculate, log, repeat.

DAY SPAN	FOCUS AREA	KEY SKILLS
1–5	REST API data ingestion	`urllib.request`, JSON parsing, error handling, exponential backoff
6–10	Technical indicators	SMA, EMA, Bollinger Bands, RSI from scratch
11–15	SQLite mastery	Schema design, `INSERT`, `SELECT`, aggregation, double-entry logging
16–20	Statistical risk metrics	Variance, standard deviation, Sharpe ratio, max drawdown
21–25	Data pipeline architecture	Batching, windowing, memory-efficient streaming
26–30	**Phase 1 Capstone**	Build a complete ingestion → calculation → logging pipeline. AST audit must pass all critical checks.

Phase 1 Score Target: 100/100 on every day by Day 30.

Common Failure Modes: - Missing try-except on network calls **(automatic 0)** - Nested O(n^2) loops on large datasets **(AST warning → -10 pts)** - Ledger imbalance (debits ≠ credits) **(test failure → -30 pts)** -

Missing type annotations on function signatures **(AST info → -10 pts)**

Phase 2: Arbitrage Logic (Days 31–60)

Theme: Find the spread. Execute before it's gone.

DAY SPAN	FOCUS AREA	KEY SKILLS
31–35	Cross-exchange detection	Poll PRIMARY vs SECONDARY, calculate spread, threshold detection
36–40	Arbitrage execution	POST `/v1/execute`, position sizing, fee-aware P&L
41–45	Triangular arbitrage	3-asset cycles (BTC→ETH→USDT→BTC), rate conversion, opportunity scanning
46–50	Risk controls	Kelly Criterion position sizing, stop-loss, max exposure limits
51–55	Latency simulation	Timestamp diff analysis, slippage models, fill probability
56–60	**Phase 2 Capstone**	Real-time triangular scanner with auto-execution, risk limits, and full ledger reconciliation

Phase 2 Score Target: 100/100 consistently. By now, failing a day should be rare.

Common Failure Modes: - Trading without checking spread threshold **(logic error → test failure)** - Executing without sufficient balance **(ledger inconsistency)** - Missing fee calculations in P&L **(SOCPA compliance violation)** - No rate limiting on exchange calls **(simulated ban → day reset)**

Phase 3: Slippage & SOCPA Compliance (Days 61–90)

Theme: It's not enough to be profitable. You must be provably correct.

DAY SPAN	FOCUS AREA	KEY SKILLS
61–65	SOCPA audit trails	Every trade has a complete provenance record, timestamps, and authorization chain
66–70	IFRS 9 reporting	Financial instrument classification, impairment calculations, disclosure notes

DAY SPAN	FOCUS AREA	KEY SKILLS
71–75	VaR & CVaR	Historical simulation VaR (95%/99%), expected shortfall, stress scenario analysis
76–80	Compliance report generation	Automated PDF/JSON reports, auditor-ready output, exception logging
81–85	System resilience	Graceful degradation, exchange downtime handling, data recovery
86–90	**Final Capstone**	Full 5-day audit simulation: run all 90 days, produce complete compliance package, pass external audit scenario

Phase 3 Score Target: 100/100. The evaluation engine becomes stricter. Partial credit is removed for any compliance violation.

Common Failure Modes: - Missing audit timestamps **(automatic 0 — SOCPA violation)** - IFRS classification errors **(test failure → -30 pts)** - VaR calculation off by more than 1% **(precision failure → day reset)** - Any ledger imbalance after reconciliation **(compliance breach)** - System crash during exchange downtime **(resilience failure)**

Weekly Schedule

Each week follows a rhythm:

DAY	ACTIVITY	ESTIMATED TIME
Monday	Receive briefing, review objectives	15 min
Tuesday	Write solution code	1–3 hours
Wednesday	Debug, run, iterate	1–2 hours
Thursday	Finalize, run EOD, fix violations	1–2 hours
Friday	Advance, review learnings	30 min
Saturday	Optional: refactor or extend previous days	–
Sunday	Rest or catch up	–

Total weekly commitment: 4–8 hours for most participants.

Scoring System

COMPONENT	WEIGHT	DETAILS
AST Critical Violations	–40 each	No try-except on network calls, eval/exec, prohibited imports. **Two = automatic 0.**
AST Warnings	–10 each	Nested loops, missing annotations, division by zero risk, infinite loops, global state
Test Failures	–30 each	Hidden test suite checks functional correctness
Ledger Imbalance	–30 each	Double-entry verification failure (debits ≠ credits)
Manual Deductions	Variable	Principal Strategist may apply situational penalties

Score = max(0, 100 – deductions)

You need **100 exactly** to advance. One warning brings you to 90. One critical brings you to 60.

Reference Materials

The simulator enforces three real-world accounting standards:

CMA (Certified Management Accountant)

- **Part 1 — Financial Planning, Performance, and Analytics**
- Section C: Financial Statement Analysis
- Section E: Investment Decisions
- **Part 2 — Strategic Financial Management**
- Section A: Financial Statement Analysis
- Section D: Risk Management

SOCPA (Saudi Organization for Chartered and Professional Accountants)

- **Standard 4:** Documentation and audit trail requirements
- **Standard 7:** Financial instrument recognition and measurement

- **Standard 12:** Revenue recognition and trade recording

IFRS (International Financial Reporting Standards)

- **IFRS 9:** Financial Instruments — classification, measurement, impairment
- **IFRS 13:** Fair Value Measurement — hierarchy levels, disclosure
- **IFRS 15:** Revenue from Contracts with Customers — trade revenue recognition

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AST Auditor — What It Checks

The AST auditor parses your code and flags these issues:

Critical (Score –40)

CODE	PATTERN	WHY IT MATTERS
`NO_NETWORK_EXCEPT`	`urllib.request.urlopen(...)` without `try`	Exchange goes down → your pipeline crashes
`NO_NETWORK_EXCEPT`	`urllib.request.Request(...)` without `try`	Same — every network call must be guarded
`EVAL_EXEC`	`eval(...)` or `exec(...)`	Arbitrary code execution — security risk
`PROHIBITED_IMPORT`	`pickle`, `shelve`, `marshal`	Deserialization vulnerabilities

Warning (Score –10)

CODE	PATTERN	WHY IT MATTERS
`NESTED_LOOP`	`for i in range(n): for j in range(m):`	$O(n^2)$ complexity — doesn't scale
`DIVISION_BY_ZERO`	`a / b` without `b != 0` check	Runtime crash
`INFINITE_WHILE`	`while True:` without `break`	Never-terminating loop
`GLOBAL_STATE`	`global x` at module level	Side effects, untestable code

Info (Score –0, but recorded)

CODE	PATTERN	WHY IT MATTERS
`MISSING_ANNOTATION`	Function parameter or return without type hint	PEP 484 compliance

Study Resources

Python Standard Library (all you need)

MODULE	USE CASE	RELEVANT DAYS
`urllib.request`	HTTP GET/POST to exchange	1–60
`urllib.error`	HTTP error handling	1–60
`json`	Parse exchange responses	1–90
`sqlite3`	Ledger database	1–90
`ast`	Static code analysis (used by auditor)	Understanding only
`subprocess`	Run tests in isolated processes	Understanding only
`threading`	Background servers (exchange + web UI)	Understanding only
`http.server`	Exchange and dashboard servers	Understanding only
`cmd`	CLI interface	Understanding only
`math`	Statistical calculations	11–30, 71–75
`statistics`	Mean, stdev for risk metrics	16–20

External Concepts (you'll need to learn)

CONCEPT	WHERE IT APPEARS	RESOURCE
SMA / EMA	Days 1–10	Investopedia: Moving Average
Bollinger Bands	Days 6–10	Investopedia: Bollinger Bands
Sharpe Ratio	Days 16–20	Wikipedia: Sharpe Ratio

CONCEPT	WHERE IT APPEARS	RESOURCE
Cross-exchange arbitrage	Days 31–35	QuantConnect: Arbitrage
Triangular arbitrage	Days 41–45	Binance Academy: Triangular Arbitrage
Kelly Criterion	Days 46–50	Investopedia: Kelly Criterion
Value at Risk (VaR)	Days 71–75	CFA Institute: VaR
CVaR / Expected Shortfall	Days 71–75	Wikipedia: Expected Shortfall

Progress Tracking

Use the simulator's built-in tracking:

```
strategist> status
```

Expected output by phase:

```
Phase 1 (Day 30):  Days passed: 30/30  Cum. score: 3000/3000  ⭐⭐⭐
Phase 2 (Day 60):  Days passed: 60/60  Cum. score: 6000/6000  ⭐⭐⭐
Phase 3 (Day 90):  Days passed: 90/90  Cum. score: 9000/9000  ⭐⭐⭐🏆
```

Graduation Criteria

To complete the Principal Strategist program:

1. **All 90 days passed** with score 100/100
2. **Zero critical AST violations** across all days
3. **Ledger balanced** at all times (verified after each day)
4. **Final capstone compliance package** generated and complete

Graduates receive: - Completion certificate (generated by the simulator on Day 90) - Their name added to the NovaCap "Honor Roll" (a JSON file in the workspace) - The satisfaction of surviving the most demanding Python training simulation ever built

"This is not a course. This is a simulation of the most demanding trading desk in the region."

Generated by NovaCap Principal Strategist — Python Standard Library Only